BICS 4340 / CSCI 4340 MACHINE LEARNING
GROUP PROJECT

Student Name & Matric No:	
Semester:	Semester II 2025/2026
Lecturer:	Prof. Dr. Amelia Ritahani binti Ismail
Due Date:	
Task Distribution (% of Work) :	Student Name (% - Total of 100% for each student)
Main Reference:	
Declaration of the Usage of Generative AI	We hereby declare that any use of Generative Artificial Intelligence (AI) tools in the preparation of this project has been conducted responsibly and transparently. The following statements apply: AI Tools Used: (Specify any AI tools or platforms used, such as ChatGPT, Grammarly, DALL·E, or others. If none were used, state "None.") Purpose of Use: (Briefly describe how AI tools were used—e.g., to improve grammar and clarity, to assist in data analysis, or to generate preliminary ideas. No content generated by AI has been directly copied without proper review and revision.) Human Oversight: All AI-assisted outputs were critically reviewed, edited, and validated by the undersigned student(s). The intellectual content, analysis, and conclusions of this work remain the result of human authorship.

<u>Learning Outcomes:</u>

1. To familiarise students with critical thinking concepts into various machine learning applications
2. To expose students to evaluate machine learning algorithms to be applied to specific computer science problems. Students can choose an algorithms that has not been studied in the classroom with a recent **combine concepts, techniques or hybrid and compare the performances of the chosen techniques with at least 3 algorithms that has been studied in class.**
3. To test **various parameter, transfer function, optimiser to the model**
4. To apply, test, and interpret machine learning algorithms together and foster a healthy intellectual discourse when completing the group project.
5. To deploy the model to be applicable to the user to be tested with new dataset.

A. Project Scope

- Projects must involve **a clear research or applied problem** suitable for machine learning modelling (e.g., classification, regression, clustering, or predictive analysis).
- Students may use **real-world datasets** (from sources such as Kaggle, UCI Repository, or institutional data) provided the data complies with ethical and privacy standards.

B. Methodological Requirements

- Students must apply **at least one algorithm not covered in class**, **integrating optimisation and hybridisation approaches**.
- A **comparative performance analysis** is required with a minimum of **three existing algorithms** taught during the semester.
- Proper **parameter tuning** and testing with various **transfer functions and optimisers** must be included.
- Evaluation metrics (e.g., accuracy, precision, recall, F1-score, RMSE, or AUC) must be reported and justified.
- deploy the model to be applicable to the user to be tested with new dataset.

C. Documentation

1. Your report should includes the following contents:

    i. Abstract
    This section explain briefly which algorithms you choose to combine, why, what challenges and brief indication of the result you achieved

    ii. Introduction
    This chapter includes the explanation on the background of your problem. Explain the data you have chosen, why you have chosen this data and what do you plan to get out of the data through machine learning.

    iii. Literature Review
    Conduct a literature review on the methods or models that you plan to combine for your project. Use this section to provide justification why you have chosen these machine learning algorithms (over others).

    iv. Data Analysis
    This chapter explain the exploratory data analysis and highlight any data pre-processing that has been done.

    v. Experimental Setup
    This chapter describes your chosen methodology. You will also need to explain here how your two algorithms will complement each other to improve model performance.

    vi. Deployment
    Explain your deployment process

    vii. Results & Discussion
    Share the results obtained from your implemented model. Discuss how the combination of the methods or algorithms chosen has affected the results of the individual algorithm or the algorithm before it's been combined. Share your opinion and theories on the results and any insight you have on how they can be improved.

    viii. Conclusion
    This section summarises your groups' work. You must explain the contribution of each group member towards completing the assignment. You need to summarise what was learnt from this entire activity and any avenues for future work and improvements.

    ix. References in APA format with a minimum of 15 articles (within 5 years)

2. After completing the project, prepare a video by all the members of the group to present your work and pitch your model. The content should include:

    i. Background
    ii. Data Analysis
    iii. Explanation of The Methodology
    iv. Discussion on the Results
    v. Conclusion and Future Work

D. Tools and Technologies

- Acceptable tools: Python, R, MATLAB, or other approved programming environments.
- Recommended libraries: TensorFlow, Keras, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib.

E. Academic Integrity

- All work must be original.
- AI tools (e.g., ChatGPT, Copilot, Grammarly) may be used **only for support**, not for content substitution.
- Any AI assistance must be transparently disclosed in the declaration.

Your Project will be marked according to the following:

1. Report
2. Data Analysis
3. Model Development
4. Experimental Process
5. Analysis of Results
6. Deployment
7. Video Group Presentation – you tube link preferred [for code / deployment explanation]
8. Individual Q&A [10 marks-individual]